

# Sujit Kumar

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## EDUCATION

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**BS in Data Science & Applications**, Indian Institute of Technology Madras May '25 — Present

**B.Tech in Electronics and Communication Engineering**, Shri Mata Vaishno Devi University, Katra Aug '25 — Present

## EXPERIENCE

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**Research Intern** Jun '26 — Jul '26

National Institute of Technology Patna [Link](#) Patna, Bihar, India

- Submitted a 28-page review report on parallel and distributed ELM neural networks under Dr. Ayaz Ahmad (NIT Patna) — six concepts, each with a uniform structure and a runnable Python program, most on real datasets
- Built ELM in Python/NumPy training in a single matrix step (Moore–Penrose pseudoinverse / ridge least-squares, no backpropagation) — 95.74% accuracy on handwritten digits, validated on breast-cancer and XOR
- Implemented distributed ELM using the MapReduce pattern, matching the single-machine result exactly ( $\sim 1e-16$ ); in a worked example, a 400 GB hidden-layer matrix reduces to  $\sim 4$  GB per machine across 100 machines
- Streamed a 54K-row dataset at near-constant memory (1.9 MB peak), measured a  $2.57\times$  speedup on 4 cores via parallel bootstrap, and ran word-count MapReduce over a 235K-word corpus

**Summer Intern** Jun '26 — Present

Afflatus Gravures Pvt. Ltd. Noida, UP, India (Remote)

- Working in the Robotics Section on Automated Defect Detection in Rotogravure Cylinders using Computer Vision

## PROJECTS

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**Python Mini-Projects (5)** [Link](#)

- Built 5 terminal apps in Python (Expense Tracker, Snake Game, Football Match Predictor, QR Code Generator, Typing Speed Test) using OOP, Pandas, Matplotlib, and qrcode; each with documented READMEs

## TECHNICAL SKILLS

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**Languages** Python, C

**Data Stack** NumPy, Pandas, Matplotlib, Scikit-learn

**Tools** Git, GitHub, MATLAB, Google Sheets, Microsoft Excel